

Quantifying the value of a potential new diagnostic test in the context of imperfect adherence

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BACKGROUND

- The **value of diagnostic tests depends** on their **adherence** and test characteristics, namely **sensitivity and specificity**.
- A high-quality standard test may not be as valuable if there is low uptake or adherence.
- New tests may have higher adherence but are not necessarily of the same quality as the standard test.
- Question:** what are the test characteristics necessary to justify the adoption of a new test?

PURPOSE

To develop a framework to determine the minimum operating criteria of a potential new diagnostic test with higher adherence than the standard test but with lower sensitivity or specificity.

QUICK FACTS

- With imperfect adherence, higher test characteristics (e.g., sensitivity and specificity) can be required to maintain the same levels of effectiveness as with perfect adherence.
- The role the new test is expected to play when adopted is relevant to identifying the minimum operation criteria.
- Our framework helps us calculate minimum operating criteria that hold the cost-effectiveness under imperfect adherence.

METHODS

Diagnostic test decision tree model

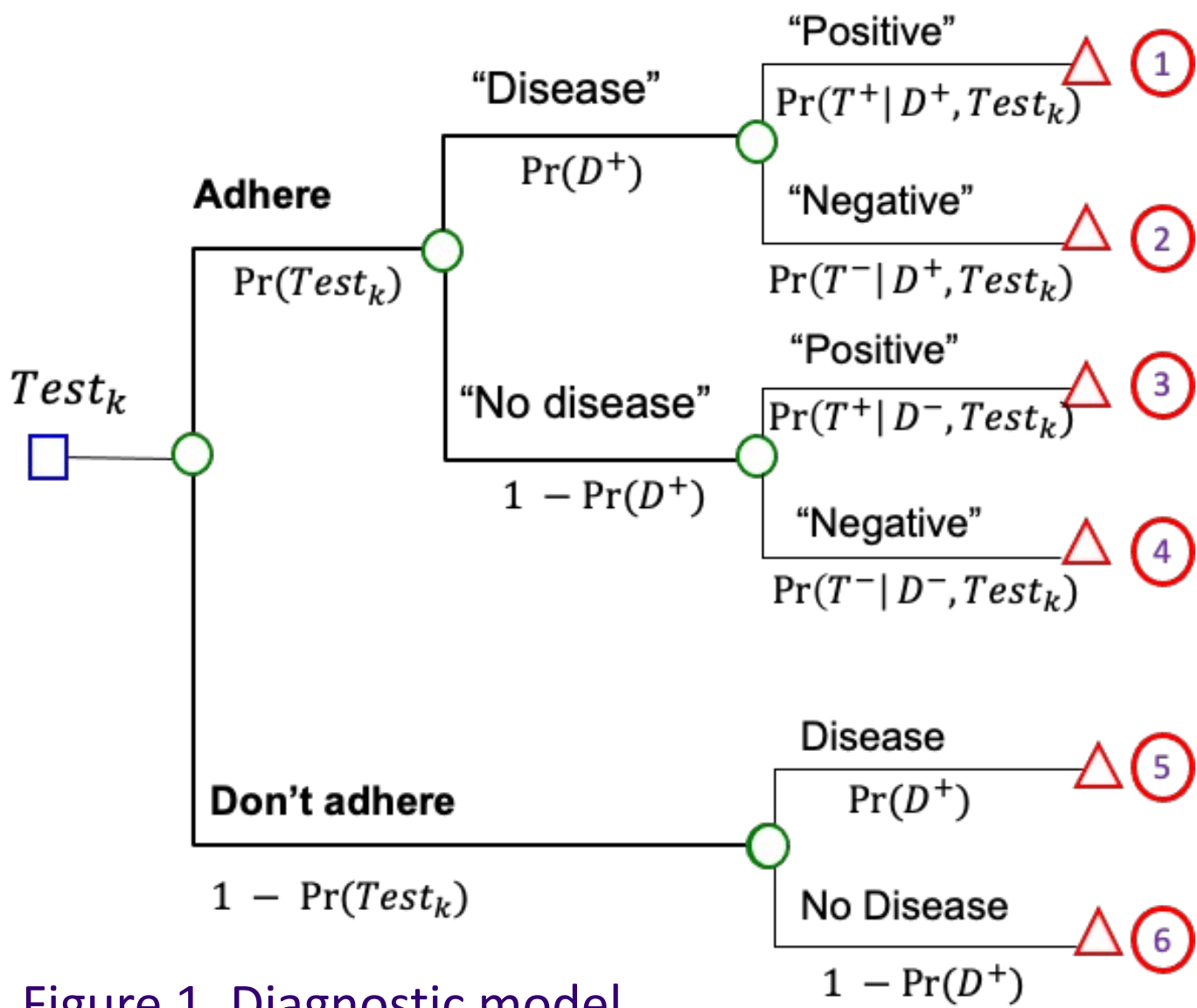


Figure 1. Diagnostic model.

- ① $NMB_{true+} = LY_{D+,Tx=1} * WTP - cost_{Test\ k} - cost_{Tx}$
- ② $NMB_{false-} = LY_{D+,Tx=0} * WTP - cost_{Test\ k} - cost_{NoTx}$
- ③ $NMB_{false+} = LY_{D-,Tx=1} * WTP - cost_{Test\ k} - cost_{InitialTx}$
- ④ $NMB_{true-} = LY_{D+,Tx=0} * WTP - cost_{Test\ k}$
- ⑤ $NMB_{not\ tested,D+} = LY_{D+,Tx=0} * WTP - cost_{NoTx}$
- ⑥ $NMB_{not\ tested,D-} = LY_{D-,Tx=0} * WTP$

Parameter	Value
$LY_{D+,T=1}$ (True positives)	0.9
$LY_{D+,T=0}$ (False negatives)	0.6
$LY_{D-,T=1}$ (False positives)	0.98
$LY_{D-,T=0}$ (True negatives)	1
WTP	100,000
$cost_{test\ 1}$	500
$cost_{test\ 2}$	500
$cost_{tx}$	250,000
$cost_{Notx}$	300,000
$cost_{Initial\ tx}$	890

Table 1. Model parameters.

Base-case scenario

We defined the reference case scenario using a current diagnostic test with the following characteristics.

Current Diagnostic test	
Sensitivity	91 %
Specificity	87 %
Adherence	60 %

Table 2. Current test characteristics

Implementation scenarios

The model will calculate the minimum sensitivity and specificity values that a new test must have in the context of imperfect adherence under three scenarios based on the role the new test is expected to play when adopted:

- **Role 1. Replacing.** The new diagnostic test is meant to substitute the current diagnostic test
- **Role 2. Complementing.** The new diagnostic test is intended to complement the current diagnostic test
- **Role 3. Switching.** A mix of both, meaning people already adherent to the current diagnostic test might switch to the new diagnostic

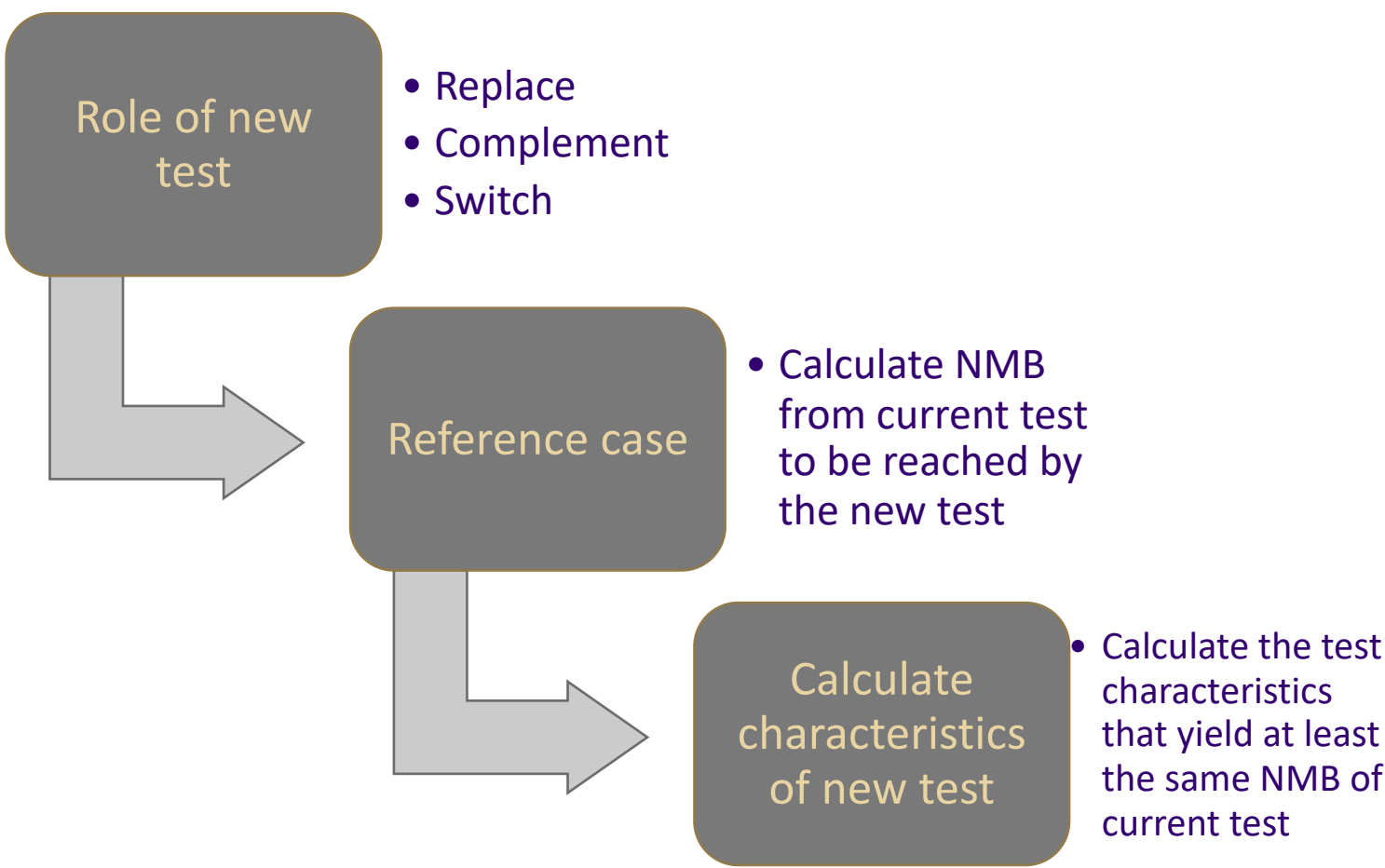


Figure 2. New test implementation scenario

Sensitivity function

From the total NMB equation, we solved for the sensitivity of the new diagnostic test to calculate the values of specificity and adherence that keep the calculated NMB constant as the reference.

$$Sensitivity_{Test_k} = f(NMB_{reference}, Specificity_{Test_k}, Pr(Test_k))$$

RESULTS

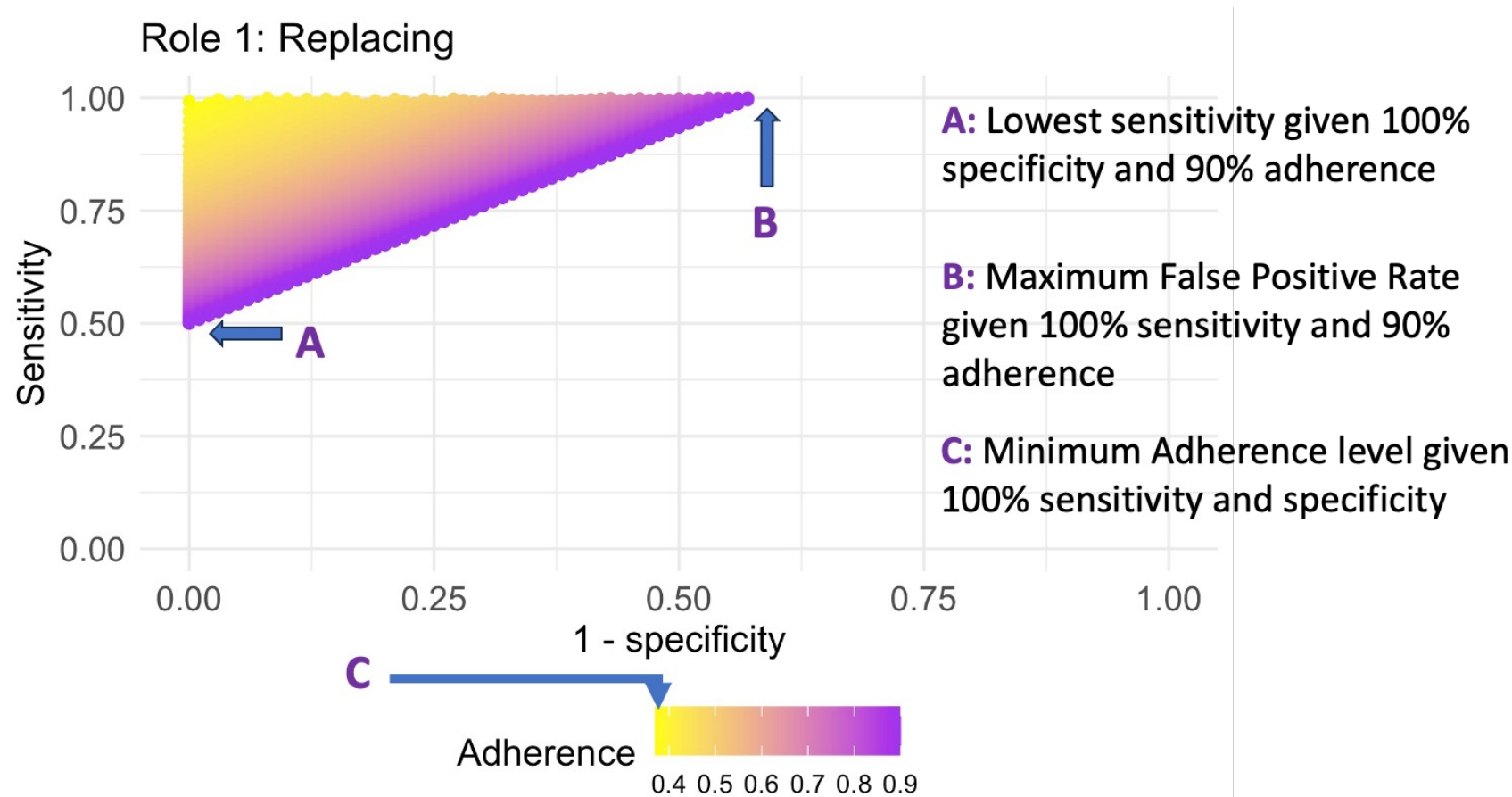


Figure 3. Interpretation of extreme values of sensitivity, specificity and adherence that the new test must have to keep the same NMB as the reference case

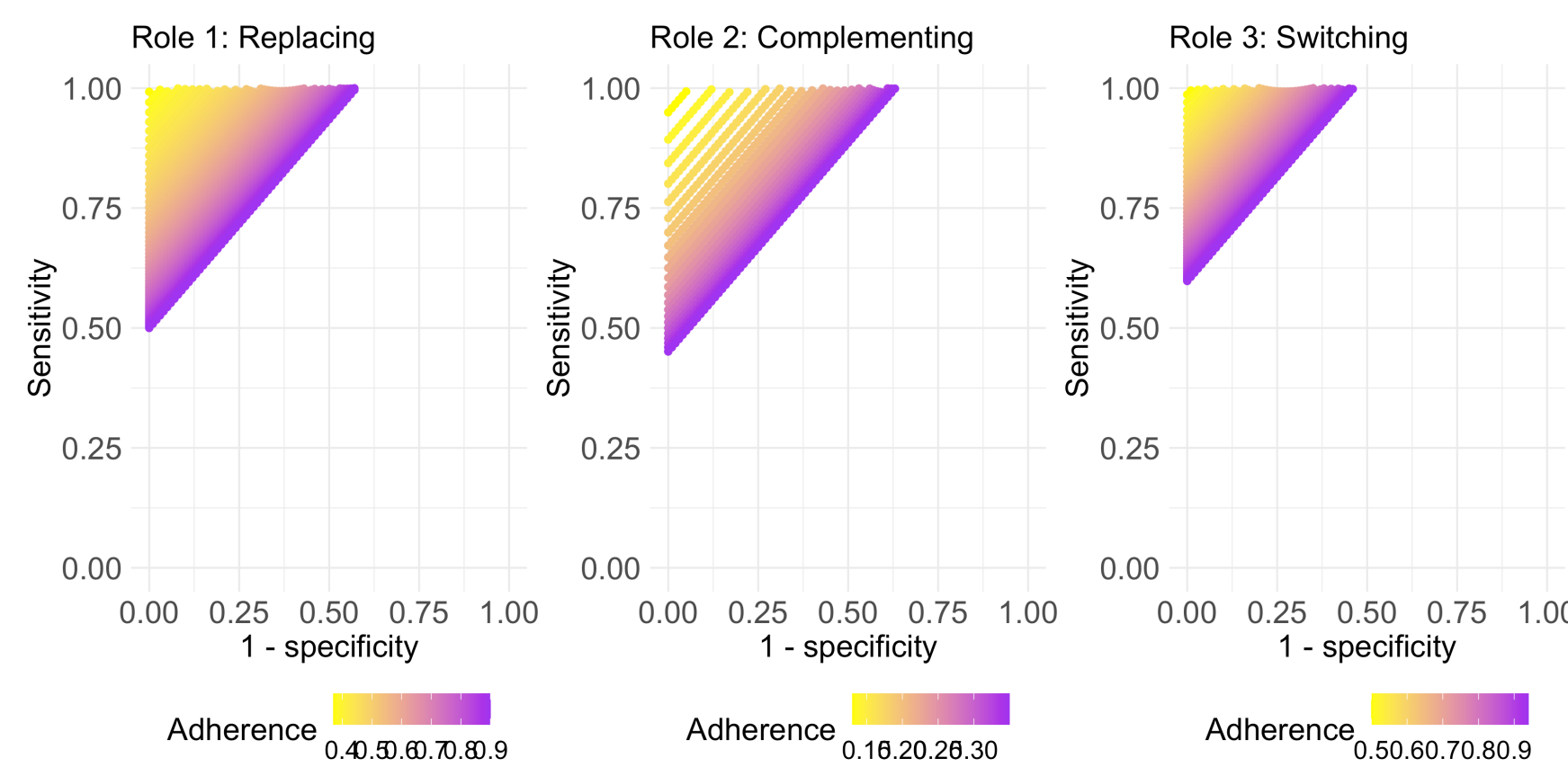


Figure 4. Sensitivity, specificity and adherence values that the new test must have to keep the same NMB as the reference case for the three roles considered.

CONCLUSIONS:

- Assuming 100% adherence overestimates the effectiveness of diagnostic tests. Consequently, poorer test characteristics are necessary to maintain the cost-effectiveness of current diagnostic tests.
- Each potential role of the new diagnostic test leads to different minimum operating criteria.
- Quantifying the value of a new test should consider potential substitutions between tests, especially when the new test has poorer sensitivity or specificity.

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